

Practical Interview Exercise

ASP.NET Core MVC CRUD Application (.NET 8)

1. Exercise Overview

Build a **Product Management web application** using **ASP.NET Core MVC** targeting **.NET 8**.

The application must support the following operations:

- View a list of products
- View product details
- Create a new product
- Edit an existing product
- Delete a product

Solution Requirements

Your implementation must include:

- SQL database schema
 - ASP.NET Core MVC frontend using Razor Views
 - Entity Framework Core for data access
 - Proper use of async/await
 - Validation and basic error handling
 - Clean folder structure and readable, maintainable code
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2. Functional Requirements

2.1 Product List

Display all active products in a table/grid format.

Required columns:

- Product Name
- SKU
- Category
- Price

- Quantity In Stock
 - Is Active
 - Created Date
 - Actions (View / Edit / Delete)
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2.2 Product Details

Display full details for a selected product.

2.3 Create Product

Provide a form to create a new product.

2.4 Edit Product

Allow users to update an existing product.

2.5 Delete Product

Allow product deletion with a confirmation step.

Note: Hard delete is acceptable. Soft delete is optional.

3. Technical Requirements

Required Technology Stack

- ASP.NET Core MVC
 - .NET 8
 - Entity Framework Core
 - SQL Server
 - Razor Views
 - Bootstrap (for UI styling)
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4. Database Schema

4.1 Categories Table

Column Name	Data Type	Constraints
Id	int	Primary Key, Identity
Name	nvarchar(100)	Required, Unique
Description	nvarchar(250)	Nullable
IsActive	bit	Required, Default = 1
CreatedUtc	datetime2	Required

4.2 Products Table

Column Name	Data Type	Constraints
Id	int	Primary Key, Identity
Name	nvarchar(150)	Required
SKU	nvarchar(50)	Required, Unique
Description	nvarchar(500)	Nullable
Price	decimal(18,2)	Required
QuantityInStock	int	Required
CategoryId	int	Required, Foreign Key (Categories.Id)
IsActive	bit	Required, Default = 1
CreatedUtc	datetime2	Required
UpdatedUtc	datetime2	Nullable

4.3 Constraints & Rules

- Category name must be unique
 - Product SKU must be unique
 - Price must be greater than 0
 - Quantity must be 0 or greater
 - Each product must belong to one category
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5. Seed Data Requirement

You must include initial data:

- Minimum **3 categories**
- Minimum **8 products**

Example Categories:

- Electronics
- Office Supplies
- Furniture

Recommended Approaches:

- EF Core seeding (**preferred**)
 - SQL script
 - Manual UI entry (least preferred)
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6. UI Requirements

Required Pages

- Products/Index
 - Products/Details
 - Products/Create
 - Products/Edit
 - Products/Delete
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Form Fields (Create/Edit)

- Product Name
- SKU
- Description
- Price
- Quantity In Stock
- Category (dropdown)
- Is Active (checkbox)

UI Expectations

- Use Bootstrap for layout and styling
- Show validation messages clearly
- Provide navigation back to list page
- Display success/error messages
- Populate category dropdown dynamically

7. Deliverables

Submit the following:

- Complete source code
- Database setup approach (EF migrations or SQL script)
- README file with setup instructions
- Any assumptions made during development

8. Final Instruction

Build a fully functional **ASP.NET Core MVC application targeting .NET 8** that manages products and categories using standard CRUD operations.

Ensure:

- Proper database design
 - Functional UI pages
 - Clean architecture
 - Asynchronous EF Core usage
 - Input validation
 - Readable, maintainable code
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